

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech. (IT) Examination****Nov 2018****IT404.01 / IT404 Language Processor****Date: 30.11.2018, Friday****Time: 01:30 p.m. To 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the following questions. [07]**

- 1) Differentiate following terms: (a) NFA and DFA (b) Syntax tree and Parse Tree 04
- 2) Identify which phase of the compiler generates an error for unbalanced parenthesis. 01
- 3) Find out different tokens from the given C statement as a part of lexical analysis. 01

$$x = a * b + 2;$$
- 4) Enumerate different types of assembly language statements. 01

Q - 2.a Discuss the role of Language Processor in program generation and program execution activities. [04]**Q - 2.b Answer any TWO questions. [10]**

- 1) What is an Assembler? Explain the functionality of Assembler.
- 2) Define three address code. Which phase of the compiler generates three address code? Write three address code for given code.

 $N = f(a[i]);$

- 3) Draw DFA for given regular expression.

 $(1 + 10 + 110)^* 0$ **Q - 3.a Apply different parameter passing methods for generating output of the given code. [02]****Begin:**

i: integer; i= 0;

procedure swap(a: integer, b: integer)

begin : a= a+ b; b= a – b; a= a-b; end

begin x[3] = {1,2,0};

swap(i, x[i]);

print(i, x[0], x[1], x[2]);

End**Q -3.b Answer any TWO questions. [12]**

- 1) Construct the precedence table for below grammar and encode the table with precedence function f() and g().

 $E \rightarrow E + E \mid E * E \mid id$

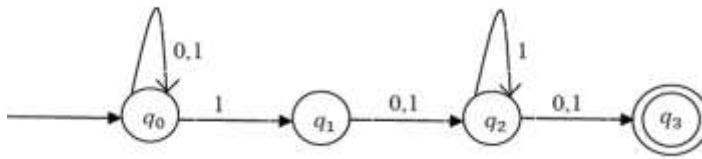
- 2) Construct NFA- $\wedge$ for following regular expression and convert it into DFA using Thompson's Construction.

 $(0^* \mid 1^*)^*$

- 3) What is the significance of Code Optimization phase in compiler design? Explain peephole optimization techniques with suitable examples.

SECTION – II**Q - 4 Do as directed. [07]**

- 1) Consider the given finite automaton. 01



Find set of reachable states for the input string 0011.

- 2) A system program that sets up an executable program in main memory ready for execution is _____. 01
- 3) Define the term: Annotated parse tree. 01
- 4) Differentiate between single-pass compiler and multi-pass compiler in brief. 02
- 5) What kind of grammar is said to be an operator precedence grammar? Check whether below grammar is an operator precedence grammar or not. 02

 $E \rightarrow EOE$ $E \rightarrow id$ $O \rightarrow + \mid *$ **Q - 5.a** What is an ambiguous grammar? Give an example of ambiguous grammar. [04]**Q - 5.b** Eliminate the left recursion from the following grammar. [05] $A \rightarrow ABd \mid Aa \mid a$ $B \rightarrow Be \mid b$ **Q - 5.c** Find out FIRST() and FOLLOW() set for following grammar. [05] $S \rightarrow ACB \mid Cbb \mid Ba$ $A \rightarrow da \mid BC$ $B \rightarrow g \mid \epsilon$ $C \rightarrow h \mid \epsilon$ **OR****Q - 5.b** Perform Left factoring on following grammar. [05] $A \rightarrow aAB \mid aBc \mid aAc$ **Q - 5.c** Construct canonical LR(1) item set and generate CLR Parsing Table for below grammar. [05] $S \rightarrow L = R \mid R$ $L \rightarrow * R \mid id$ $R \rightarrow L$ **Q - 6.a** Discuss the role of symbol table in compiler design. Also explain any one data structure which is suitable for implementation of symbol table with an example. [05]**Q - 6.b** Attempt Any THREE questions. [09]

- 1) What is activation record? Draw and explain the structure of an activation record.
- 2) Explain “shift-reduce” and “reduce-reduce” conflict with suitable example.
- 3) Explain the input buffering technique with example.
- 4) List and explain the different types of grammar in detail.
